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Number Systems



The odometer is a device used to measure the speed of a vehicle, while the speedometer is a device used to measure the distance travelled by a vehicle. An irrational number, pi (π), is used by engineers in many tasks every day, including calibrating these instruments in vehicles.



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TOPIC 1

NUMBER LINE

Learning Objectives

- ➔ Students will be learning about various types of numbers, viz., natural numbers, whole numbers, prime numbers, rational numbers, etc., which constitute the number system.

Learning Outcomes

- ➔ Students will be able to explore the relation among various types of numbers.
- ➔ Students will be able to find the rational numbers between two given rational numbers.

Real Life Application

How do you measure your body temperature when you are suffering from a high fever and your body temperature rises? We all use thermometers for that.



In the regular thermometer, the length of the liquid column, i.e., mercury varies depending on the temperature. When you use a thermometer to check your body temperature, you must have noticed the long hash marks beneath which numbers are written in Celsius and Fahrenheit, so you can see a number line on the thermometer where various numbers are written at equal intervals.

So, when the numbers are represented at equal distances on a line we called it a number line.

Now, let us gain more knowledge about the number line and study various types of numbers in detail.

Natural Numbers

Numbers (that are part of the number system) that include all positive integers from 1 to infinity are called natural numbers. They are most commonly used for counting. A collection of natural numbers is denoted by **N**.

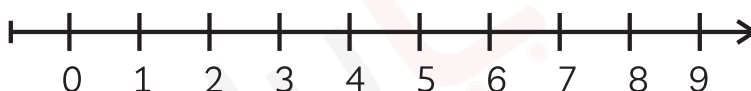


Important

- 1 is the smallest natural number.
- The largest natural number cannot be found.

Whole Numbers

If we include zero (0) in a group of all natural numbers, a new group of numbers is formed, called whole numbers. Therefore, the collection of all natural numbers including zero is known as whole numbers and it is denoted by **W**.



Important

- Zero is the smallest whole number.
- There is no largest whole number as there are infinite whole numbers.

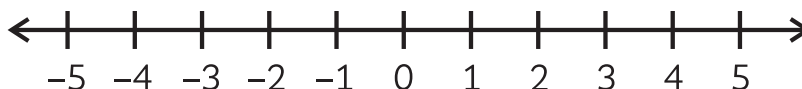


Caution

- Students should know that every natural number is a whole number but every whole number need not to be a natural number.

Integers

A collection of whole numbers with all positive and negative whole numbers and no decimal or fractional part, is called integers. The collection of integers is denoted by **Z**.



Important

- Integers less than zero are known as negative integers, and integers greater than zero are known as positive integers.
- The symbol **Z** comes from the German word 'Zahlen', which means 'to count'.



Rational Numbers

Numbers which can be written in the form of $\frac{p}{q}$, where p, q are integers

and $q \neq 0$, are known as rational numbers. The collection of rational numbers is denoted by **Q**.

The word 'Rational' comes from the word 'ratio' and **Q** comes from the word 'quotient'.

For example, -35 can be written as $\frac{-35}{1}$ here, $p = -35$ and $q = 1 \neq 0$.

Therefore, the rational numbers also include all natural numbers, whole numbers and integers.



Important

→ Every integer is a rational number.

Equivalent Rational Numbers

The rational numbers have a unique representation in the form $\frac{p}{q}$,

where p and q are integers and $q \neq 0$.

For example, $\frac{1}{2} = \frac{2}{4} = \frac{10}{20} = \frac{50}{100}$

These are equivalent rational numbers (or fractions). Equivalent rational numbers can also be found as equivalent fractions by multiplying or dividing the same number to the numerator and the denominator of the given rational number.

Methods of Finding Rational Numbers Between Two Numbers

Let a and b be two rational numbers, such that $a > b$.

Then, $\frac{a+b}{2}$ is a rational number exactly lying between a and b .



Important

→ There are infinite rational numbers between two given rational numbers.

Example 1. Find five rational numbers between $\frac{3}{5}$ and $\frac{4}{5}$. [NCERT]

Ans. Given, rational numbers are $\frac{3}{5}$ and $\frac{4}{5}$.

Write the equivalent fractions of the given fractions, by multiplying numerator and denominator with $(5 + 1)$, we get,

$$\frac{3}{5} \times \frac{6}{6} = \frac{18}{30}$$

$$\frac{4}{5} \times \frac{6}{6} = \frac{24}{30}$$

∴ Thus, the five required rational numbers between $\frac{3}{5}$ and $\frac{4}{5}$ are

$$\frac{19}{30}, \frac{20}{30}, \frac{21}{30}, \frac{22}{30}, \frac{23}{30}$$

Caution

→ If you want n rational numbers, then convert the given fractions into their equivalent fraction by multiplying numerator and denominator by $(n + 1)$.

If a and b are the two rational numbers such that $b > a$, then we can also find n rational numbers between them as

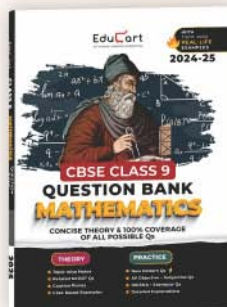
$(a + d), (a + 2d), (a + 3d), \dots, (a + nd)$, where $d = \frac{b-a}{n+1}$.

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•
•



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OBJECTIVE Type Questions

Multiple Choice Questions

[1 mark each]

1. The value of $1.\overline{79}$ in $\frac{p}{q}$ form, where p and q are integers, is :

(a) $\frac{179}{99}$

(b) $\frac{178}{99}$

(c) $\frac{176}{99}$

(d) $\frac{170}{99}$

Ans. (b) $\frac{178}{99}$

Explanation: Let $x = 1.\overline{79}$... (i)

Multiply equation (i) by 100, we get

$$100x = 179.79...$$

$$100x = 178 + 1.79...$$

$$100x = 178 + x$$

[From eq. (i)]

$$99x = 178$$

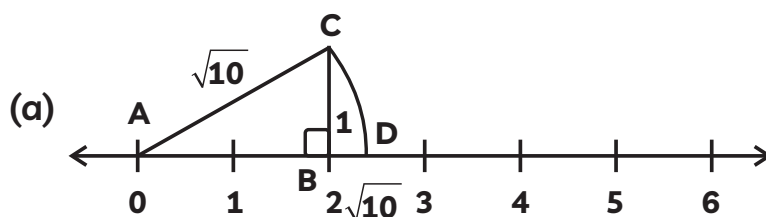
$$x = \frac{178}{99}$$



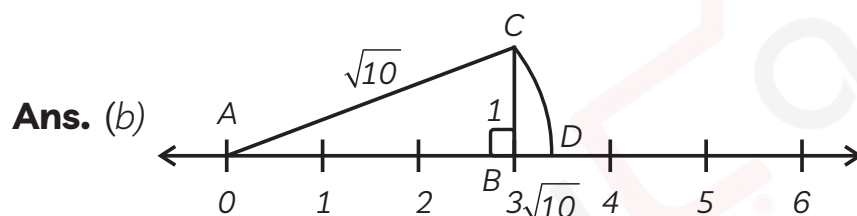
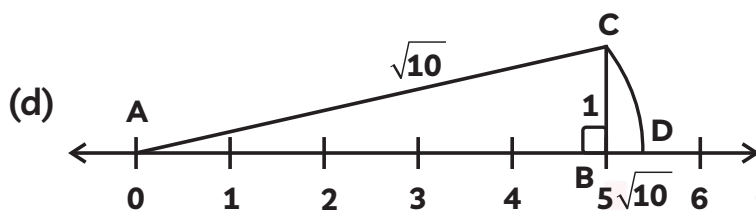
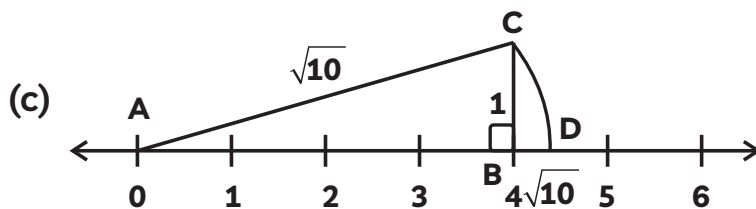
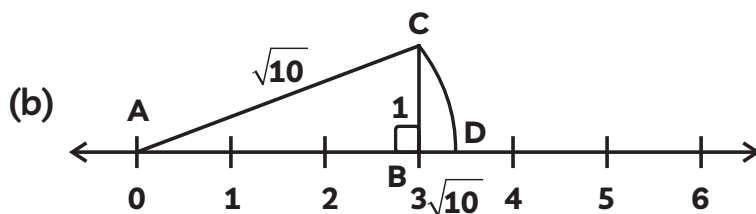
Caution

- Students usually get confused while identifying the number of digits that should be multiplied. So remember that the multiplication depends upon the number of digits containing the bar. If 2 digits contain a bar multiply by 100, if four digits contain a bar then by 10000.

2.* Which of the following represents $\sqrt{10}$ on a number line?



* Topics and Questions which are a part of latest CBSE Syllabus but have been removed by NCERT.



Explanation:

- (1) Take $AB = 3$ unit on a number line. (Consider 1 unit = 1 cm)
- (2) At point B, draw a perpendicular, and mark C such that $BC = 1$ unit.
- (3) Now, keeping A as the center and AC as radius, draw an arc to cut the number line at D.

Distance AD will show the value of $\sqrt{10}$.

In right angle triangle ABC, by Pythagoras Theorem,

$$AC^2 - AB^2 = BC^2$$

$$AC^2 - (3)^2 = (1)^2$$

$$AC = \sqrt{9+1} = \sqrt{10}$$

True and False

[1 mark each]

3. $\frac{\sqrt{18}}{\sqrt{2}}$ is not a rational number as $\sqrt{18}$ and $\sqrt{2}$ are not integers.

Ans. False



Explanation: Here, $\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$

Hence, $\frac{\sqrt{18}}{\sqrt{2}} = \frac{3\sqrt{2}}{\sqrt{2}} = 3$; which is a rational number .

Fill in the Blanks

[1 mark each]

Fill in the blanks with suitable information:

4. A rational number lying in between is given by

Ans. $\frac{a+b}{2}$

Explanation: $\frac{a+b}{2}$ will be the number between a and b .

Assertion-Reason

[A-R]

[1 mark each]

In the following question, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
- (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
- (c) Assertion (A) is true but reason (R) is false.
- (d) Assertion (A) is false but reason (R) is true.

5. Assertion (A): The sum of two irrational numbers $2 + \sqrt{5}$, and $1 + \sqrt{3}$ is an irrational number.

Reason (R): The sum of irrational numbers is always an irrational number.

Ans. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

Explanation: Given,

$$(2 + \sqrt{5}) + (1 + \sqrt{3}) = 3 + \sqrt{5} + \sqrt{3}$$

Hence, it is an irrational number.

CASE BASED Questions (CBQs)

[5 marks each]

Read the following passages and answer the questions that follow:

6. As schools are closed due to the lockdown, children in Rural Jharkhand are unable to attend online classes due to a lack of access to mobile phones. So Mr. Rajesh Kumar started a mobile classroom and teaches Mathematics and Science to the students. One day while teaching "Number Systems" he discuss about the rational numbers and their decimal expansions. He reiterate the class that a number that can be written in numbers in the form of $\frac{p}{q}$, where p and q both are integers and $q \neq 0$ and can be expanded into terminating or non-terminating decimals depending upon the number. He writes two numbers $\frac{3}{11}$ and $0.\overline{58}$ on the whiteboard and ask some question to the students.



(A) The decimal form of $\frac{3}{11}$ is:

- (a) 0.27 (b) 0.27
(c) 0.27 (d) 0.27

(B) Identify the $\frac{p}{q}$ form of $0.\overline{58}$, where p and q both are integers and $q \neq 0$, as:

- (a) $\frac{53}{90}$ (b) $\frac{58}{90}$
(c) $\frac{58}{99}$ (d) $\frac{53}{99}$

(C) The sum of $\frac{3}{11}$ and $0.\overline{58}$ is a/an:

- (a) whole number (b) integer
(c) rational number (d) irrational number

Ans. (A) (b) 0.27

Explanation:

$$\begin{array}{r} 11 \overline{) 3000} \quad 0.2727 \\ \underline{22} \\ 80 \\ \underline{77} \\ 30 \\ \underline{22} \\ 80 \\ \underline{77} \\ 3 \end{array}$$

$$\therefore \frac{3}{11} = 0.272727\ldots = 0.\overline{27}$$

(B) (c) $\frac{58}{99}$

Explanation: Let $x = 0.\overline{58} = 0.5858\ldots$... (i)

Multiply both sides by 100, we get

$$100x = 58.5858 \ldots \text{ (ii)}$$

Subtract eq. (i) from eq. (ii), we get

$$\begin{aligned} 99x &= (58.5858...) - (0.5858...) \\ &= 58.000... \end{aligned}$$

$$99x = 58$$

$$x = \frac{58}{99}$$

(C) *rational number*

Explanation: Here,

$$0.\overline{58} = \frac{58}{99}.$$

$$\text{Therefore, } \frac{3}{11} + \frac{58}{99} = \frac{27+58}{99} = \frac{85}{99},$$

it is a rational number.

VERY SHORT ANSWER Type Questions (VSA)

[1 mark each]

7. Find the value of $(144)^{0.50} \times (625)^{0.25}$.

Ans. $(144)^{0.50} \times (625)^{0.25}$

$$\begin{aligned} &= (144)^{\frac{1}{2}} \times (625)^{\frac{1}{4}} \\ &= \left[(12)^2\right]^{\frac{1}{2}} \times \left[5^4\right]^{\frac{1}{4}} \\ &= (12)^{2 \times \frac{1}{2}} \times (5)^{4 \times \frac{1}{4}} \\ &= 12 \times 5 = 60. \end{aligned}$$

SHORT ANSWER Type-I Questions (SA-I)

[2 marks each]

8. Evaluate:

$$(A) \frac{2^{38} + 2^{37} + 2^{36}}{2^{39} + 2^{38} + 2^{37}}$$

$$(B) \left[\left(64^{\frac{1}{2}} \right)^{\frac{1}{6}} \right]^2$$

[Delhi Gov. QB 2022]



Ans. (A) $\frac{2^{38} + 2^{37} + 2^{36}}{2^{39} + 2^{38} + 2^{37}}$

$$= \frac{2^{36} \times 2^2 + 2^{36} \times 2^1 + 2^{36}}{2^{37} \times 2^2 + 2^{37} \times 2^1 + 2^{37}}$$

$$= \frac{2^{36}(2^2 + 2^1 + 1)}{2^{37}(2^2 + 2^1 + 1)}$$

$$= \frac{2^{36}}{2^{36} \times 2^1} = \frac{1}{2}$$

(B) $\left[\left(64^{\frac{1}{2}} \right)^{\frac{1}{6}} \right]^2 = 2^{6 \times \frac{1}{2} \times \frac{1}{6} \times 2} = 2$

SHORT ANSWER Type-II Questions (SA-II)

[3 marks each]

9. An expression is given: $2(\sqrt{k} - 1) + \sqrt{8}$. If we add $-8\sqrt{2}$ to the expression, the result so obtained is a rational number, then what is the value of k ?

Ans. Given expression is $2(\sqrt{k} - 1) + \sqrt{8}$

Now, add $-8\sqrt{2}$ to the given expression, we get

$$= 2(\sqrt{k} - 1) + \sqrt{8} - 8\sqrt{2}$$

$$= 2(\sqrt{k} - 1) + 2\sqrt{2} - 8\sqrt{2}$$

$$= 2(\sqrt{k} - 1) - 6\sqrt{2}$$

$$= 2\sqrt{k} - 2 - 6\sqrt{2}$$

Since, the result obtained will be a rational number. We have to eliminate $6\sqrt{2}$ from the expression,

$$\begin{aligned}\text{Thus,} \quad & 2\sqrt{k} = 6\sqrt{2} \\ \text{or} \quad & \sqrt{k} = 3\sqrt{2} \\ \text{or} \quad & k = 3^2 \times 2 \\ \text{or} \quad & k = 18\end{aligned}$$

Thus, the value of k is 18.

LONG ANSWER Type Questions (LA)

[4 & 5 marks each]

10. If $A = 0.36363636\ldots$ and $B = 0.636363\ldots$, then what is the

value of $\frac{1}{A} + \frac{1}{B}$?

Ans. If $A = 0.363636\ldots$
 $A = 0.\overline{36}$... (i)

Multiply by 100

$$100A = 36.\overline{36} \quad \dots \text{(ii)}$$

Subtract eq. (i) from eq. (ii)

$$100A - A = 36.\overline{36} - 0.\overline{36}$$

$$99A = 36$$

$$A = \frac{36}{99}$$

$$A = \frac{4}{11}$$

$$B = 0.636363\ldots$$

If $B = 0.\overline{63}$... (iii)

Multiply by 100

$$100B = 63.\overline{63} \quad \dots \text{(iv)}$$

Subtract eq. (iii) from eq. (iv)

$$100B - B = 63.\overline{63} - 0.\overline{63}$$

$$99B = 63$$

$$B = \frac{63}{99}$$

$$B = \frac{7}{11}$$

Now,

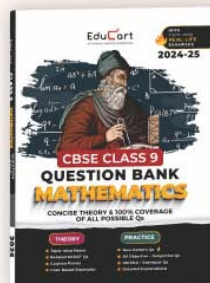
$$\frac{1}{A} + \frac{1}{B} = \frac{11}{4} + \frac{11}{7}$$

$$= \frac{77 + 44}{28} = \frac{121}{28}$$



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